Study Material 1

Read Chapters 1-2 of Database System Concepts and answer the following questions.

1. Fill in the blanks.

(a) Underlying the structure of a database is the (data model): a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.

(b) In the relational model, data are represented in the form of (tables or relations). A table has multiple (columns, attributes, variables, or features); Each column has a unique (name). Each (row, record, data instance, or data row) of the table represents one piece of information.

(c) The collection of information stored in the database at a particular moment is called a/an (instance of the database or database instance). The overall design of the database is called the (database schema).

(d) A database system provides a/an (data-definition language or DDL) to specify the database schema and a/an (data-manipulation language or DML) to express database queries and updates. In practice, these are not two separate languages; instead, they simply form parts of a single language.

(e) A (query) is a statement requesting the retrieval of information.

(f) A/an (Transaction manager) ensures that the database remains in a consistent state despite system failures, and that concurrent transaction executions proceed without conflicts.

(g) A/an (File manager) manages the allocation of space on disk storage and the data structures used to represent information stored on disk.

(h) A/an (Buffer manager) is responsible for fetching data from disk storage into main memory, and deciding what data to cache in main memory.

(i) The (null value) is a special value that signifies the value is unknown or does not exist.

2. What are the major disadvantages of keeping organizational information in a file-processing system?

pp. 5-8

- Data redundancy and inconsistency
- Difficulty in accessing data
- Data isolation
- Integrity problems
- Atomicity problems
- Concurrent-access anomalies
- Security problems

3. (Exercise problem 1.7) List four significant differences between a file-processing system and a DBMS.

- Both systems manage collections of data and provide mechanisms to access that data. However, a DBMS controls both the physical storage and the logical organization of the data, whereas a file-processing system typically manages only how the data is physically stored and accessed.
- A DBMS helps reduce data redundancy by maintaining a centralized and consistent representation of data that can be shared among multiple applications. In contrast, file-processing systems often lead to duplicated data because different programs store and manage their own copies.

- DBMSs allow users to access data in a flexible and declarative way, such as through queries. File-processing systems generally require predefined programs to retrieve or manipulate data.
- A DBMS is designed to support concurrent access by multiple users to the same dataset. In file-processing systems, multiple programs can usually access the same file simultaneously only if they perform read-only operations.

4. (Exercise problems 2.6, 2.8, and 2.14)
Consider the employee database of Figure 2.17 (page 60). Give an **expression in the relational algebra** to express each of the following queries:
a. Find the name of each employee who lives in city "Miami".

$$\Pi_{person_name}(\sigma_{city = \text{"Miami"}}(employee))$$

b. Find the name of each employee whose salary is greater than $100,000. *Note that the salary refers to the yearly wage*.

$$\Pi_{person_name}(\sigma_{salary > 100000}(employee \bowtie works))$$

c. Find the name of each employee who lives in "Miami" and whose salary is greater than $100,000.

$$\Pi_{person_name}(\sigma_{city = \text{"Miami"} \wedge salary > 100000}(employee \bowtie works))$$

Now consider the employee database below and give an **expression in the relational algebra** to express each of the following queries:

employee (*ID, person_name, street, city*)
works (*person_name, company_name, salary*)
company (*company_name, city*)

d. Find the ID and name of each employee who does not work for "BigBank".

Possible sol 1: $\Pi_{ID,\ person_name}(\sigma_{company_name <> \text{"BigBank"}}(employee \bowtie_{employee.person_name = works.\ person_name} works))$

Possible sol 2: $\Pi_{ID,\ person_name}(employee)$
$- \Pi_{ID,\ person_name}(employee \bowtie_{employee.person_name = works.\ person_name}(\sigma_{company_name = \text{"BigBank"}}(works)))$

e. Find the ID and name of each employee who earns at least as much as every employee in the database.

$$\Pi_{ID,\ person_name}(employee)$$
$$- \Pi_{A.ID,\ A.person_name}(\rho_A(employee \bowtie works) \bowtie_{A.salary < B.salary} \rho_B(employee \bowtie works))$$

f. Find the ID, name, and city of residence of each employee who works for “BigBank”.

$$\Pi_{ID,\ person_name,\ city}\ (employee \bowtie_{employee.person_name\ =\ works.person_name}\ (\sigma_{company_name\ =\ \text{“BigBank”}}\ (works))$$

g. Find the ID, name, street address, and city of residence of each employee who works for “BigBank” and earns more than $10,000.

$$\Pi_{ID,\ person_name,\ street,\ city}\ (\sigma_{(company_name\ =\ \text{“BigBank”}\ \wedge\ salary\ >\ 10000)}(\ works \bowtie_{employee.person_name=works.person_name}\ employee))$$

h. Find the ID and name of each employee in this database who lives in the same city as the company for which s/he works.

$$\Pi_{ID,\ person_name}$$
$$(\sigma_{employee.city=company.city}(employee \bowtie_{employee.person_name=works.person_name}\ works$$
$$\bowtie_{works.company_name=company.company_name}\ company))$$

5. (Exercise problem 2.18) Write the following **queries in relational algebra**, using the university schema (Figure 2.8, page 46).

a. Find the ID and name of each instructor in the Physics department.

$$\Pi_{ID,name}(\sigma_{dept_name=\text{”Physics"}}\ (instructor))$$

b. Find the ID and name of each instructor in a department located in the building “Watson”.

$$\Pi_{ID,name}(instructor \bowtie_{instructor.dept_name=department.dept_name}\ (\sigma_{building=\text{“Watson"}}\ (department)))$$

c. Find the ID and name of each student who has taken at least one course in the “Comp. Sci.” department.

$$\Pi_{ID,name}(student \bowtie_{student.ID=takes.ID}\ takes \bowtie_{takes.course_id=course.course_id}\ \sigma_{dept_name=\text{“Comp. Sci."}}\ (course))$$

d. Find the ID and name of each student who has taken at least one course section in the year 2018.

$$\Pi_{ID,name}(student \bowtie_{student.ID=takes.ID}\ \sigma_{year=2018}\ (takes))$$

e. Find the ID and name of each student who has not taken any course section in the year 2018.

$$\Pi_{ID,name}(student) - \Pi_{ID,name}(student \bowtie_{student.ID=takes.ID}\ (\sigma_{year=2018}\ (takes)))$$

6. Warming up with MySQL. In Week 3, the connection information for ***mayo***, the DBMS used in this course, will be provided. Once you receive the instructions, follow them to connect to the database using the database client installed on your machine. Then explore the provided databases and try executing SQL queries to become familiar with the relational database and SQL environment.
Below examples use the "*university*" database (NOT *university_small*). This database shares the same schema as the one used in the lectures but contains a larger set of data records collected during a different period of time.

a. List the names of all databases that the *mayo* DBMS has.

Answer:

classicmodels
employees
foodmart
sakila
university
university_small
world
world_x

SQL Query to obtain your answer:

```
SHOW DATABASES;
```

b. List the names of all tables that the "university" database has.

Answer:

advisor
classroom
course
department
instructor
prereq
section
student
takes
teaches
time_slot

SQL Query to obtain your answer:

```
USE university;
SHOW TABLES;
```

c. Execute and explain the differences among the results of the following queries.

(i) `SELECT * FROM instructor;`
(ii) `SELECT 'Teacher' FROM instructor;`
(iii) `SELECT 'Teacher';`
(iv) `SELECT *, 'Teacher' FROM instructor;`

⇨ Possible answers

(i) returns all rows and columns of the instructor table
(ii) returns the 'Teacher' literals (strings) as much as the number of instructors (50 times)

(iii) returns a ‘Teacher’ literal
(iv) returns all the instructor table with an additional columns of the ‘Teacher’ literals appended to the right